

Data Cleaning Documentation

3.1 Data Profiling Before Cleaning

Total Row & Column:

- A. Master Data Product
Row: 20
Column Attribute: 10
- B. Master Data Vendor
Row: 3
Column Attribute: 7
- C. Master Data Patner Store
Row: 50
Column Attribute: 6
- D. Master Data Sales Team
Row: 5
Column Attribute: 5
- E. Data Sales Order
Row: 1028
Column Attribute: 7
- F. Data Order Item
Row: 3392
Column Attribute: 5
- G. Data Invoice
Row: 1028
Column Attribute: 8

Primary Key & Foreign Key Identification:

- A. Master Data Product
Primary Key: ProductID
Foreign Key: VendorID
- B. Master Data Vendor
Primary Key: VendorID
Foreign Key: -
- C. Master Data Patner Store
Primary Key: PatnerStoreID
Foreign Key: -
- D. Master Data Sales Team
Primary Key: SalesTeamID
Foreign Key: -

E. Data Sales Order

Primary Key: SalesOrderID

Foreign Key: PatnerStoreID, SalesTeamID,

F. Data Order Item

Primary Key: -

Foreign Key: SalesOrderID, ProductID

G. Data Invoice

Primary Key: InvoiceID

Foreign Key: SalesOrderID

Statistic Summary for numerical values data before data cleaning

A. Master Data Product

Metric	UnitSalesPrice	UnitBuyPrice	Quantity	DiscountProduct
Count	20	20	20	20
Mean	1141295	967100	6766,35	21%
Median	323000	262500	3455	20%
Standard Deviation	1483812,75	1269865,758	5412,01631	6%
Minimum	20000	14000	1418	10%
Maximum	5639000	4800000	17743	30%
25% Percentile	217250	178750	3007	19%
75% Percentile	2236000	1838750	11179,25	25%

B. Master Data Vendor

-

C. Master Data Patner Store

-

D. Master Data Sales Team

Metric	TotalProspect
Count	5
Mean	10
Median	10
Standard Deviation	4,242640687
Minimum	6
Maximum	16
25% Percentile	6
75% Percentile	12

E. Data Sales Order

Metric	TotalOrderTransaction
Count	1028
Mean	Rp.286.195.461,79
Median	Rp.228.568.870,00
Standard Deviation	Rp.262.108.450,39
Minimum	Rp.918.000,00
Maximum	Rp.1.548.353.450,00
25% Percentile	Rp.74.238.852,50
75% Percentile	Rp.409.400.247,75

F. Data Order Item

Metric	QuantityOrder	ItemTotal
Count	3392	3392
Mean	87,73761792	Rp.86.736.124,62
Median	88	Rp.26.875.850,00
Standard Deviation	36,14031308	Rp.125.662.819,61
Minimum	25	Rp.405.000,00
Maximum	150	Rp.795.099.000,00
25% Percentile	57	Rp.12.613.518,50
75% Percentile	119	Rp.111.991.000,00

G. Data Invoice

Metric	ShippingCost	TotalDue
Count	1028	1028
Mean	541380,6654	Rp.318.218.343,25
Median	808800	Rp.254.234.435,85
Standard Deviation	368219,2397	Rp.290.951.751,10
Minimum	8088	Rp.1.878.330,00
Maximum	859350	Rp.1.719.228.379,50
25% Percentile	32352	Rp.83.251.838,78
75% Percentile	859350	Rp.454.487.483,90

Data Interpretation(Statistical Summary) before data cleaning

A. Master Data Product

A.1 UnitSalesprice

- Skewed Distribution

1141295 (Mean) > 323000 (Median)

Result = Right Skewed

This meaning many of data(majority) has small sales price, but there's some value data has high amount of Unit Sales Price.

- Outlier (using method IQR & boundary, then compare to Max metric)

IQR

75% Percentile – 25% Percentile

$2236000(75\% \text{ Percentile}) - 217250(25\% \text{ Percentile})$
 2118750
 Lower Bound
 $217250(25\% \text{ Percentile}) - (1,5 * 2118750(IQR))$
 -2960875
 Upper Bound
 $2236000(75\% \text{ Percentile}) + (1,5 * 2118750(IQR))$
 5514125

Result: $(5639000) \text{ Max} > (5514125) \text{ Upper bound}$. This meaning that there's outlier for this Unit Sales Price data. Also, this mean there's data value that is significantly different or extreme compared to most of the other data.

- Spread Data
 $1483812,75 \text{ (Standard Deviation)} > 1141295 \text{ (Mean)}$
 This meaning this data so highly variable

Standard deviation that is higher than the mean indicates a high level of data variability, suggesting the presence of potential outliers or an uneven distribution of prices. Therefore, further analysis was conducted to detect and validate extreme values.

- Data Anomaly
 - Upper bound and Max metric has similar value, that mean there's Unit Sales Price data has high value than others
 - Median = 323000 against Mean = 1141295, that huge gap between this metric. There's indication uneven data or data not distribute equally. If we deep understanding of this data, many products have cheap Unit Sales Price data and only little product have expensive Unit Sales Price compare all Unit Sales Price data.

A.2 UnitBuyPrice

- Skewed Distribution
 $967100 \text{ (Mean)} > 262500 \text{ (Median)}$
 Result = Right Skewed
 This meaning many of data(majority) has small Buy price, but there's some value data has high amount of Unit Buy Price.
- Outlier (using method IQR & boundary, then compare to Max metric)
 IQR
 $75\% \text{ Percentile} - 25\% \text{ Percentile}$
 $1838750(75\% \text{ Percentile}) - 178750(25\% \text{ Percentile})$
 1660000
 Lower Bound

$178750 \text{ (25\% Percentile)} - (1,5 * 1660000 \text{ (IQR)})$
 -2311250
 Upper Bound
 $1838750 \text{ (75\% Percentile)} + (1,5 * 1660000 \text{ (IQR)})$
 4328750

Result: $(4800000) \text{ Max} > (4328750) \text{ Upper bound}$. This meaning that there's outlier for this UnitBuyPrice data. Also, this mean there's data value that is slightly different or extreme compared to most of the other data.

- Spread Data
 $1269865,758 \text{ (Standard Deviation)} > 967100 \text{ (Mean)}$
 This meaning this data so highly variable

Standard deviation that is higher than the mean indicates a high level of data variability, suggesting the presence of potential outliers or an uneven distribution of prices. Therefore, further analysis was conducted to detect and validate extreme values.

- Data Anomaly
 $\text{Max (4800000) exceeds upper bound (4328750)} \rightarrow \text{indicates outliers}$
 $\text{Mean (967100)} \gg \text{Median (262500)} \rightarrow \text{uneven / right-skewed distribution}$
 $\text{Standard Deviation (1269865,758)} > \text{Mean} \rightarrow \text{high data variability}$
 $\text{Wide range Min - Max (14000 - 4800000)} \rightarrow \text{large gap between low and high price products}$

A.3 Quantity

- Skewed Distribution
 $6766,35 \text{ (Mean)} > 3455 \text{ (Median)}$
 Result = Right Skewed
 This meaning many of data(majority) has small Quantity, but there's some value data has high Quantity value.
- Outlier (using method IQR & boundary, then compare to Max metric)
 IQR
 $75\% \text{ Percentile} - 25\% \text{ Percentile}$
 $11179,25 \text{ (75\% Percentile)} - 3007 \text{ (25\% Percentile)}$
 $8172,25$
 Lower Bound
 $3007 \text{ (25\% Percentile)} - (1,5 * 8172,25 \text{ (IQR)})$
 $-9251,375$
 Upper Bound
 $11179,25 \text{ (75\% Percentile)} + (1,5 * 8172,25 \text{ (IQR)})$
 $23437,625$

Result: (17743) Max < (23437,625) Upper bound. This meaning there isn't no outlier based on this method.

- Spread Data
5412,01631 (Standard Deviation) < 6766,35 (Mean)
Std (5,412) is lower than mean (6,766) → indicates moderate variability
Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
Max (**17,743**) < upper bound (~**23,437**) → no outlier detected
Mean (**6,766**) > Median (**3,455**) → right-skewed distribution
Std (**5,412**) < Mean → moderate variability
Range (**1,418 – 17,743**) → relatively wide but still reasonable

A.4 DiscountProduct

- Skewed Distribution
21% (Mean) > 20% (Median)
Slight right-skew (mean > median)
Discount values are relatively consistent, with occasional higher discounts (25%–30%).
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile
25%(75% Percentile) – 19% (25% Percentile)
6%
Lower Bound
19% (25% Percentile) – (1,5 * 6% (IQR))
10%
Upper Bound
25%(75% Percentile) + (1,5 * 6% (IQR))
34%

Result: (30%) Max < (34%) Upper bound. This meaning there isn't no outlier based on this method.

- Spread Data
6% (Standard Deviation) < 21% (Mean)

Std (6%) is lower than mean (21%) → indicates moderate variability
Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
Mean (**21%**) > Median (**20%**) → slightly right-skewed distribution
Std (**6%**) is relatively low → low variability (data stable)
Range (**10% – 30%**) → within reasonable and controlled range

No outlier detected → all values fall within expected boundaries

D. Master Data Sales Team

D.1 TotalProspect

- Skewed Distribution
10 (Mean) = 10 (Median)
Result = Symmetrical Distribution
This meaning data indicates balanced and evenly distributed data.
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile
12(75% Percentile) – 6 (25% Percentile)
6
Lower Bound
6 (25% Percentile) – (1,5 * 2118750 (IQR))
-3
Upper Bound
12(75% Percentile) + (1,5 * 2118750 (IQR))
21

Result: (16) Max < (21) Upper bound. This meaning there isn't no outlier based on this method.
- Spread Data
4,242640687 (Standard Deviation) < 10 (Mean)
→ indicates moderate variability. Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
No skewness (mean = median)
Moderate variability (std low)
No outlier detected

E. Data Sales Order

E.1 TotalOrderTransaction

- Skewed Distribution
Rp.286.195.461,79 (Mean) > Rp. 228.568.870,00 (Median)
Right Skewed
Indicates most transactions are lower in value, with a few high-value transactions pulling the average upward
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile

Rp. 409.400.247,75 (75% Percentile) – Rp. 74.238.852,50 (25% Percentile)
335.161.395,25

Lower Bound

Rp. 74.238.852,50 (25% Percentile) – (1,5 * 335.161.395,25(IQR))
-428.503.240,375

Upper Bound

12(75% Percentile) + (1,5 * 2118750 (IQR))
912.142.340,625

Result: (1.548.353.450) Max > (912.142.340,625) Upper bound.

Max (**Rp1.548M**) > Upper Bound (~**Rp912M**) → outlier detected

Indicates presence of extreme high-value transactions beyond normal range

- Spread Data

262.108.450,39 (Standard Deviation) < 286.195.461,79 (Mean)

→ indicates moderate variability. Suggests the data is fairly stable and not widely dispersed.

- Data Anomaly

- The transaction distribution is uneven (dominated by lower-value transactions)

- There are very high-value transactions (outliers) that influence the average

- Overall, the pattern is still reasonable within a business context (bulk / high-value orders)

F. Data Order Item

F.1 QuantityOrder

- Skewed Distribution

87,73761792 (Mean) < 88 (Median)

Left Skewed

This mean many data are low-quantity orders (might be as test orders, samples, emergency small buys, or new customers) lowering the overall average of all quantity order

- Outlier (using method IQR & boundary, then compare to Max metric)

IQR

75% Percentile – 25% Percentile

119 (75% Percentile) – 57 (25% Percentile)

62

Lower Bound

57 (25% Percentile) – (1,5 * 62(IQR))

-36

Upper Bound

119(75% Percentile) + (1,5 * 62 (IQR))

212

Result: (150) Max < (212) Upper bound.

This meaning there isn't no outlier based on this method.

- Spread Data
36,14031308 (Standard Deviation) < 87,73761792 (Mean)
→ indicates moderate variability. Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
Mean (**87.7**) < Median (**88**) → left skewness (distribution slightly show many of data has low value of Quantity Order)
No outlier detected (max **150** < upper bound ~**212**)
Moderate variability (std **36.1** < mean) → data fairly stable
Range (**25 – 150**) → relatively wide but still within reasonable limits

F.2 ItemTotal

- Skewed Distribution
Rp.86.736.124,62 (Mean) > 26.875.850,00 (Median)
Right Skewed
Indicates most item total is lower in value, with a few high-value item total pulling the average upward
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile
Rp.111.991.000,00 (75% Percentile) – 12.613.518,50 (25% Percentile)
Rp.99.377.481,50

Lower Bound
Rp.12.613.518,50 (25% Percentile) – (1,5 * Rp.99.377.481,50(IQR))
Rp.-136.452.703,75

Upper Bound
Rp.111.991.000,00(75% Percentile) + (1,5 * Rp.99.377.481,50(IQR))
Rp.261.057.222,25

Result: (Rp.795.099.000,00) Max > (Rp.261.057.222,25) Upper bound.
With this method, outlier detected. This indicates presence of extreme high-value Item Total beyond normal range.
- Spread Data
Rp.125.662.819,61 (Standard Deviation) > Rp.86.736.124,62 (Mean)
→ Indicates data is widely dispersed with significant differences between transactions

- Data Anomaly
Mean (**Rp86.7M**) >> Median (**Rp26.8M**) → right-skewed distribution
Outlier detected (max **Rp795M** > upper bound ~**Rp261M**) → extreme high-value transactions
High variability (std **Rp125M** > mean) → data widely dispersed
Wide range (**Rp405K – Rp795M**) → large gap between low and high transaction values

G. Data Invoice

G.1 ShippingCost

- Skewed Distribution
541380,6654 (Mean) < 808800 (Median)
Left Skewed
Indicates most shipping costs are higher, with a few very low values pulling the average downward
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile
859350 (75% Percentile) – 32352 (25% Percentile)
826998
Lower Bound
 $32352 (25\% \text{ Percentile}) - (1,5 * 826998(IQR))$
-1208145
Upper Bound
 $859350 (75\% \text{ Percentile}) + (1,5 * 826998(IQR))$
2099847

Result: (859350) Max < (2099847) Upper bound.
This meaning there isn't no outlier based on this method.
- Spread Data
368219,2397 (Standard Deviation) < 541380,6654 (Mean)
→ indicates moderate variability. Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
Mean (**541K**) < Median (**808K**) → left-skewed distribution
No upper outlier (max **859K** ≈ upper bound) → values capped at high end
Low-value anomaly (min **8K** << Q1 **32K**) → presence of unusually low shipping costs
Moderate variability (std **368K** < mean) → relatively stable data
Narrow upper range (**75%Percentile = Max = 859K**) → indicates possible pricing cap / *flat rate*

G.2 TotalDue

- Skewed Distribution
Rp.318.218.343,25 (Mean) > Rp.254.234.435,85 (Median)
Right Skewed
Indicates most item total is lower in value, with a few high-value item total pulling the average upward
- Outlier (using method IQR & boundary, then compare to Max metric)
IQR
75% Percentile – 25% Percentile
Rp.454.487.483,90 (75% Percentile) – Rp.83.251.838,78 (25% Percentile)
Rp.371.235.645,12
Lower Bound
Rp.83.251.838,78 (25% Percentile) – (1,5 * Rp.371.235.645,12(IQR))
Rp.-473.601.629,68
Upper Bound
Rp.454.487.483,90 (75% Percentile) + (1,5 * Rp.371.235.645,12 (IQR))
Rp.1.011.340.951,58

Result: (Rp.1.719.228.379,50) Max > (Rp.1.011.340.951,58) Upper bound.
This meaning there isn't no outlier based on this method.
- Spread Data
Rp.290.951.751,10 (Standard Deviation) < Rp.318.218.343,25 (Mean)
→ indicates moderate variability. Suggests the data is fairly stable and not widely dispersed.
- Data Anomaly
Mean (**Rp318M**) > Median (**Rp254M**) → right-skewed distribution
Outlier detected (max **Rp1.719M** > upper bound ~**Rp1.01B**) → extreme high-value transactions
Wide range (**Rp1.8M** – **Rp1.7B**) → significant gap between low and high transaction values

3.2 Data Validation

Data validation was conducted in three layers: basic validation to ensure correct data types and formats, business rule validation to verify logical consistency (e.g., price and quantity must be positive), and relational integrity checks to ensure valid relationships between tables through foreign key validation.

- Basic Validation

- Master Data Product
ProductID and VendorID classified as key and varchar data.
ProductName and ProductDescription classified as text data

Category and UOM data classified as categorical (nominal) data and text data.
UnitSalesPrice(float),UnitBuyPrice(float),Quantity(integer),
DiscountProduct(Decimal) classified as Numeric data

- Master Data Vendor
VendorID classified as key and varchar data.
VendorName, Address, City, Province, Email, Phone classified as text data
- Master Data Patner Store
PatnerStoreID classified as key and varchar data
PatnerStoreName, PatnerType(Categorical), Region, Province, and
PatnerContactPhone classified as text data
- Master Data Sales Team
SalesTeamID classified as key and varchar data
SalesTeamName, SalesTeamRegion , SalesPromotionMethods as text data
TotalProspect classified as numerical
- Data Sales Order
SalesOrderID, PatnerStoreID, SalesTeamID classified as key and varchar data
OrderDate classified as Date data
TotalOrderTransaction classified as Numerical data
- Data Order Item
SalesOrderID, ProductID classified as key and varchar data
QuantityOrder, ItemTotal classified as numerical data
- Data Invoice
InvoiceID, SalesOrderID classified as key and varchar data
ShipmentType, WarehouseLocation, TermsOfPayment classified as text data
ShippingCost, Taxes(Percentage), TotalDue classified as numerical data

- Business Rules Validation

- Master Data Product
 - Category must be within predefined valid values:
(Fishing Equipment, Billiard, Golf, Outdoors Sport, Hiking & Camping,
Bicycle, Basketball, Futsal, Gym & Fitness)
 - UnitSalesPrice data must have higher value than UnitBuyPrice for every
product
 - The quantity for each product must be greater than 0, as all products are
recorded based on actual sales transactions. The amount of every quantity is
last stock record data.
 - For data DiscountProduct is default discount for each product..
 - Based on stakeholder input, some products may not have a description.

- Master Data Partner Store
 - PartnerType is classified into two categories: Distributor and Retail.
 - Region are classified into predefined categories, including Bali, Java, Kalimantan, Maluku, North Maluku, Nusa Tenggara, Sulawesi, and Sumatera.
 - Each province is mapped to its corresponding region.
- Master Data Sales Team
 - Each sales team is assigned to a specific region but may share similar sales promotion methods
 - Each sales team must have more than 6 prospects. The aggregated total prospect count (50) is consistent with the total number of partner stores (50), indicating full coverage.
- Data Sales Order
 - Order date must be value as historical date. So there isn't Future date for OrderDate data
 - Payment is classified into two categories: Cash and Credit.
 - TotalOrderTransaction is calculated as the sum of all ItemTotal values (from the Order Item table) within a single Sales Order.
 - Based on stakeholder input, some Sales Order may not have a note.
- Data Order Item
 - QuantityOrder must greater than > 0 .
 - ItemTotal calculation :

$$(\text{UnitSalesPrice}(\text{from Master data Product}) * \text{QuantityOrder}) * (1 - \text{DiscountProduct}(\text{Percentage value}))$$
- Data Invoice
 - ShipmentType is classified into two categories: Cargo and Regular.
 - WarehouseLocation is refer to data Province(Table Partner Store)
 - TermsOfPayment data is classified into categories : CBD, COD, EOM, NET60, NET90
 - ShippingCost is determined based on the distance between the partner store location and the warehouse location.
 - Taxes is flat rate.
 - TotalDue Calculation :

$$(\text{TotalOrderTransaction}(\text{from table Sales Order}) + \text{Taxes}) + \text{Shipping Cost}.$$
- Relational Integrity
 - Master Data Product
 1. ProductID is primary key
 2. VendorID is foreign key refer relation to table Vendor.

- Master Data Vendor
 1. VendorID is primary key
 - Master Data Partner Store
 1. PartnerStoreID is primary key
 - Master Data Sales Team
 1. SalesTeamID is primary key
 - Data Sales Order
 1. SalesOrderID is primary key
 2. PartnerStoreID is foreign key refer relation to table Partner Store
 3. SalesTeamID is foreign key refer relation to table Sales Team
 - Data Order Item
 1. SalesOrderID is foreign key refer relation to table Data Sales Order
 2. ProductID is foreign key refer relation to table Master Data Product
 3. This table does not have a primary key; therefore, a **composite key** is created using **SalesOrderID and ProductID** as the unique identifier
- Note: “Uniqueness of the composite key (SalesOrderID + ProductID) was validated to prevent duplicate records.”
- Data Invoice
 1. InvoiceID is primary key
 2. SalesOrderID is foreign key refer relation to table Data Sales Order

3.3 Data Cleaning Approach Step

- Master Data Product
 1. Check if there's data duplicate for ProductID and VendorID, validate and confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 2. Check if there's data duplicate for ProductName data, validate and confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 3. Ensure both data Unit Sales Price and Unit Buy Price greater than 0.
 4. Ensure for every product Unit Buy Price data greater than Unit Sales Price data
 5. Standardize price format data UnitSalesPrice and UnitBuyPrice from general to currency numeric format. This step applied to make data ready to analyze.
 6. Ensure every data quantity value greater than 0.
 7. Change format data quantity from general to numeric format.

- Master Data Vendor
 1. Check if there's data duplicate for VendorID, validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 2. Check if there's data duplicate for VendorName, validate and confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 3. Standardize province values by change Province Indonesia to Jakarta on field Province.
 4. Standardize Phone data by change format to consistent formal telephone number(For the example : +62 xxxx xxxx).

- Master Data Patner Store
 1. Check if there's data duplicate for PatnerStoreID, validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 2. Check if there's data duplicate for PatnerStoreName, validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 3. Standardize abbreviated province values to their corresponding full and official province names.
 4. Standardize PatnerContactPhone data by change format to consistent formal telephone number(For the example : +62 xxxx xxxx).

- Master Data Sales Team
 1. Check if there's data duplicate for SalesTeamID, validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 2. Check if there's data duplicate for SalesTeamName, validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 3. Standardize abbreviated SalesTeamName values to their corresponding full and official region names(For example : from Sales Sumkal Region Team to be Sales Sumatera -Kalimantan Region Team)
 4. Change format data TotalProspect from general to numeric format.

- Data Sales Order
 1. Check if there's data duplicate for SalesOrderID validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 2. Check if there's data PatnerStoreID and SalesTeamID align with master data Patner Store and master data Sales Team.
 3. Change format data OrderDate from General format to date format.
 4. OrderDate values are filtered to retain only records before 2023.
 5. Check data validation for data Payment should be between CASH or CREDIT.
 6. Change Format Data TotalOrderTranscation from general format to numeric currency format.
 7. Check every records data TotalOrderTransaction value need to be match with data ItemTotal in table Order Item for every sales order data. If not match or tally, so need recalculate again.

- Data Order Item
 1. Check if there's data SalesOrderID and ProductID align with data in table Sales Order and master data Product.
 2. Set Create a composite key name as “*OrderItemID*” to ensure each record is unique and prevent duplicates
 3. Check if there's data duplicate for OrderItemID validate, confirm the correct records with stakeholders. Otherwise, proceed to the next step.
 4. If there is duplicate OrderItemID or any change to the ProductID, the ItemTotal data must be recalculated.
 5. For data QuantityOrder change format to numeric format.
 6. For data ItemTotal change format to Numeric Currency format.
 7. For data OrderItemID change format to text format.
- Data Invoice
 1. Check data InvoiceID and SalesOrderID if there's duplicate or not. Also check data SalesOrderID match with InvoiceID.
 2. For WarehouseLocation data, standardize the values by converting province names to their full, clear forms—for example, change “Kalsel” to “South Kalimantan”.
 3. Change ShippingCost and TotalDue format to Numeric Currency format.
 4. If the TotalOrderTransaction value in the SalesOrder table changes, the TotalDue must be recalculated.
 5. Check data taxes to ensure that tax data is consistent.

3.4 Data Post Cleaning Testing

- Data Validation Consistency Check(For numeric calculation data)

ProductID	VendorID	ProductName	UnitSalesPrice	UnitBuyPrice	Margin per Product (UnitSalesPrice - UnitBuyerPrice)
PD0001	VD0002	1 Set Fishing Equipment	Rp249.000,00	Rp180.000,00	Rp69.000,00
PD0002	VD0002	Fishing soft lure	Rp20.000,00	Rp14.000,00	Rp6.000,00
PD0003	VD0003	Stick Billiard	Rp931.000,00	Rp785.000,00	Rp146.000,00
PD0004	VD0003	Stick Billiard Bag	Rp220.000,00	Rp175.000,00	Rp45.000,00
PD0005	VD0001	Stick Golf Wedge	Rp2.195.000,00	Rp1.785.000,00	Rp410.000,00
PD0006	VD0001	1 set Golf Ball	Rp209.000,00	Rp195.000,00	Rp14.000,00
PD0007	VD0001	Stick Golf Bag	Rp3.200.000,00	Rp2.800.000,00	Rp400.000,00
PD0008	VD0002	Sunglasses	Rp149.000,00	Rp118.000,00	Rp31.000,00
PD0009	VD0002	Tracking Pole	Rp129.000,00	Rp105.000,00	Rp24.000,00
PD0010	VD0002	Hiking Shoes	Rp325.000,00	Rp250.000,00	Rp75.000,00
PD0011	VD0003	Mountine Bike	Rp2.359.000,00	Rp2.000.000,00	Rp359.000,00
PD0012	VD0003	Road Bike	Rp5.639.000,00	Rp4.800.000,00	Rp839.000,00
PD0013	VD0001	1 Set Basketball	Rp479.000,00	Rp395.000,00	Rp84.000,00
PD0014	VD0001	1 Set Futsalball	Rp267.000,00	Rp225.000,00	Rp42.000,00
PD0015	VD0002	Abs Roller	Rp135.900,00	Rp95.000,00	Rp40.900,00
PD0016	VD0002	Pull Up Bar	Rp267.000,00	Rp210.000,00	Rp57.000,00
PD0017	VD0002	Bench Press	Rp2.650.000,00	Rp2.200.000,00	Rp450.000,00
PD0018	VD0002	Dumbell Set	Rp532.000,00	Rp485.000,00	Rp47.000,00
PD0019	VD0001	Treadmill	Rp2.549.000,00	Rp2.250.000,00	Rp299.000,00
PD0020	VD0001	Heavy bag	Rp321.000,00	Rp275.000,00	Rp46.000,00

A. Master Data Product

Margin per product = UnitSalesPrice – UnitBuyPrice. This objective to know if there's product will get negative price value if use for sales

From data Margin per product checking, negative values were found in the calculations.

B. Data Sales Order and Data Order Item

$\text{Sum}(\text{TotalOrderTransaction}) = \text{Sum}(\text{TotalItem})$

Rp296.769.900.803,00 = Rp296.769.900.803,00

C. Data Invoice

Check calculation for data TotalDue.

Data TotalDue cleaning = Rp329.971.129.215,33

Data TotalDue Check Calculation Testing = Rp329.971.129.215,33

Result : consistent

- Relation table check & test :

A. Master data Product

After use Vlookup for data vendorID from Master data table Vendor, all data now cross-referenced and valid.

B. Data Sales Order

After Vlookup for data PatnerStoreID from Master data table Patner Store and SalesTeamID from Master data table Sales Team, all data now cross - referenced and valid.

C. Data Order Item

After Vlookup for data SalesOrderID from Data Sales Order and ProductID from Master Data Product, all data now cross – referenced and valid.

D. Data Invoice

After Vlookup, for data SalesOrderID from data Sales Order. All data now cross – referenced and valid.

- Data Quality check from Visualization

1. Outlier data

1.A Master Data Product

- Data field UnitSalesPrice

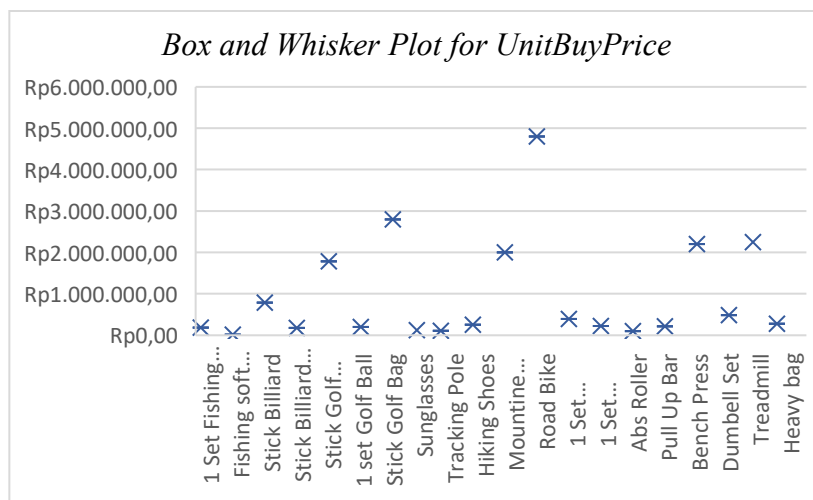


Box and Whisker Plot for UnitSalesPrice

Explanation :

From this chart, we found data product Road Bike has most significant outlier with unit sales price Rp.6.000.000. This value beyond average sales price for other product. But for this data already confirm by stakeholders that the data sales price for Road Bike was right.

- Data field UnitBuyPrice

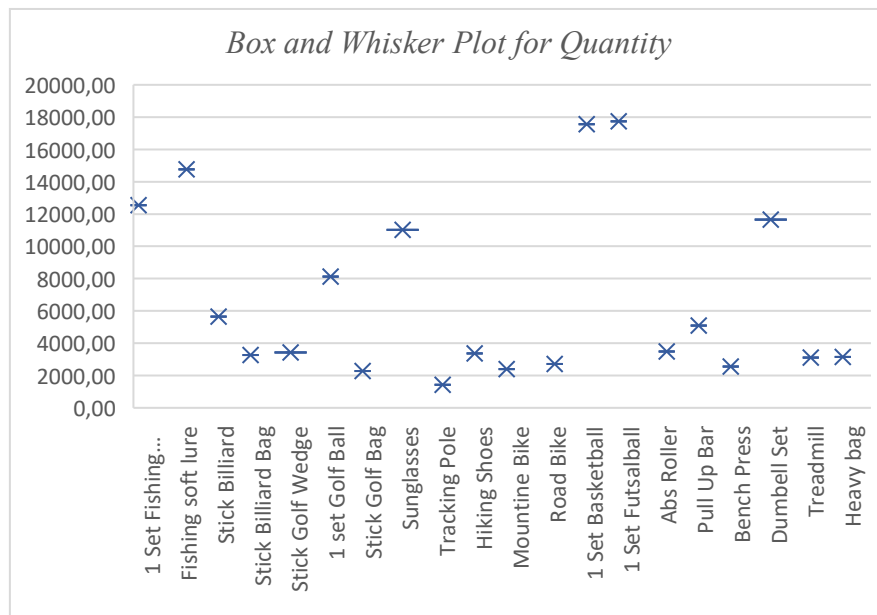


Box and Whisker Plot for UnitBuyPrice

Explanation :

From this chart, we found data product Road Bike has most significant outlier with unit buy price approximately Rp.4.000.000 – Rp.5.000.000. This value beyond average buy price for other product. But for this data already confirm by stakeholders that the data buy price for Road Bike was right.

- Data field Quantity

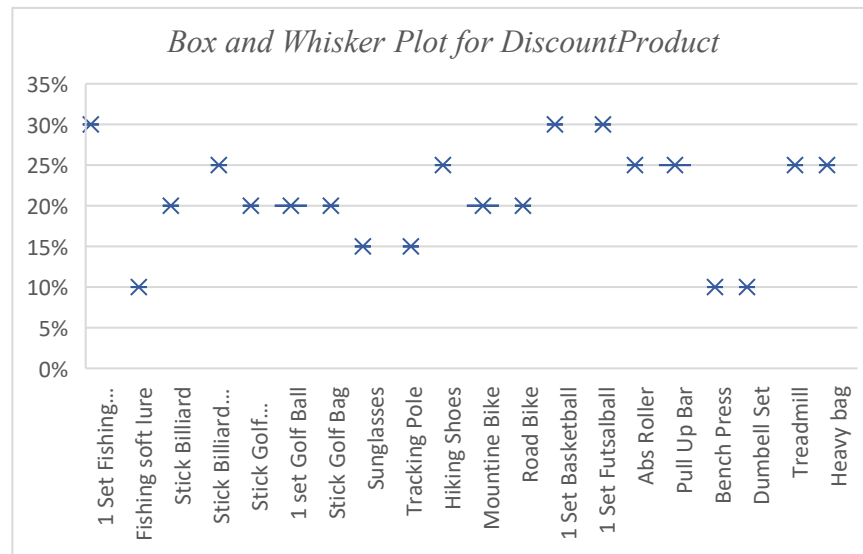


Box and Whisker Plot for Quantity

Explanation :

Although no outliers are detected using the IQR method(from statistic summary), the presence of high-value quantities suggests potential business-driven variations, such as bulk stock or high-demand products.

- Data field DiscountProduct



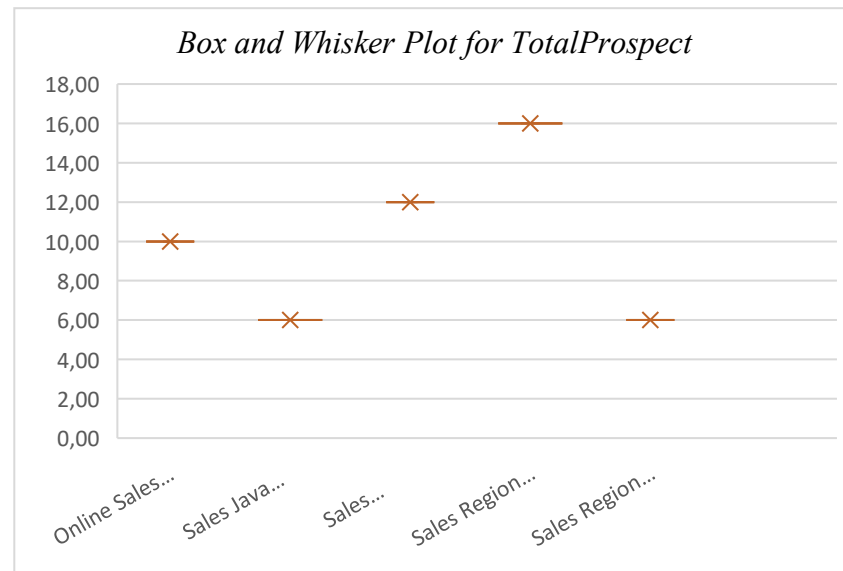
Box and Whisker Plot for DiscountProduct

Explanation :

DiscountProduct data is consistently distributed within a controlled range (10%–30%) with no detected outliers, indicating stable and standardized discount for each product.

1.B Master Data Sales Team

- Data field TotalProspect



Box and Whisker Plot for TotalProspect

Explanation :

TotalProspect data is evenly distributed with no detected outliers, indicating balanced performance across sales teams.

1.C Data Sales Order

- Data field TotalOrderTransaction



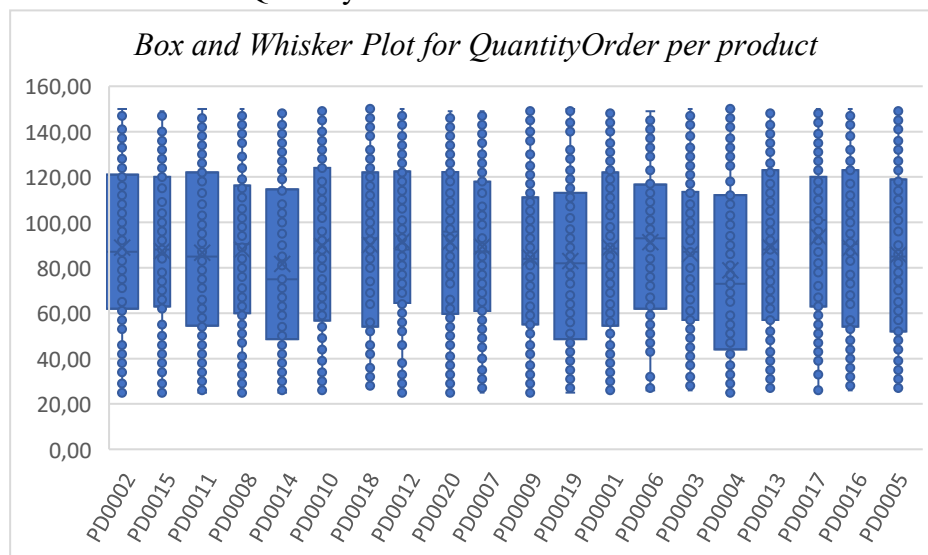
Box and Whisker Plot for TotalOrderTransaction

Explanation :

Most of the transactions happen between Rp.80.000.000 - Rp.415.000.000. The upper whisker and the many outliers indicate that while your floor(minimum) is consistent, the max still volatile(inconsistent).

1.D Data Order Item

- Data field QuantityOrder



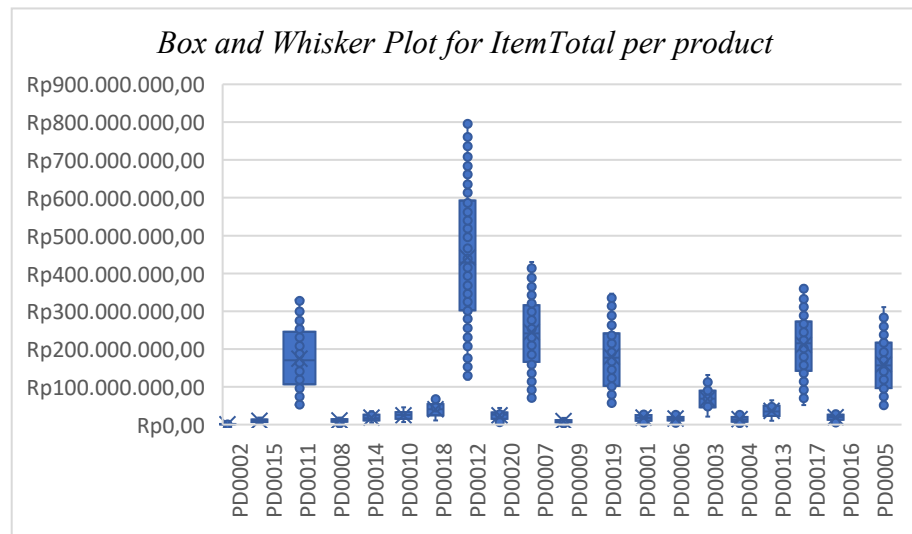
Box and Whisker Plot for QuantityOrder

Explanation :

The boxplot shows the distribution of QuantityOrder across different products. Overall, the data appears relatively consistent, with most products having a similar range of order quantities. No significant

outliers are observed, indicating that order quantities are generally within a controlled and expected range across products.

- Data field ItemTotal

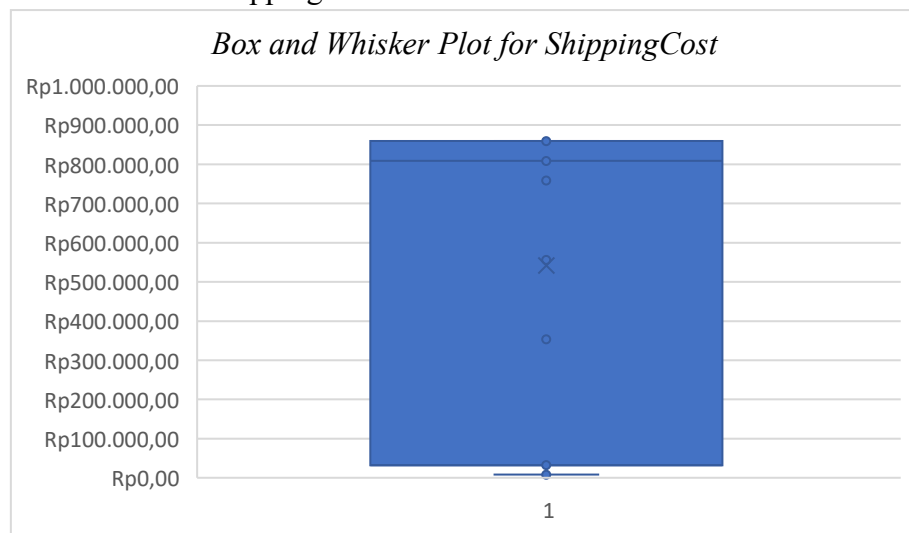


Explanation :

Presence of high-value outliers suggests that certain products are occasionally purchased in large quantities or bulk orders, significantly impacting total transaction value. Variation in spread across products suggests that majority products have more stable pricing and demand, while others are influenced by irregular or bulk purchasing behaviors.

1.E Data Invoice

- Data field ShippingCost

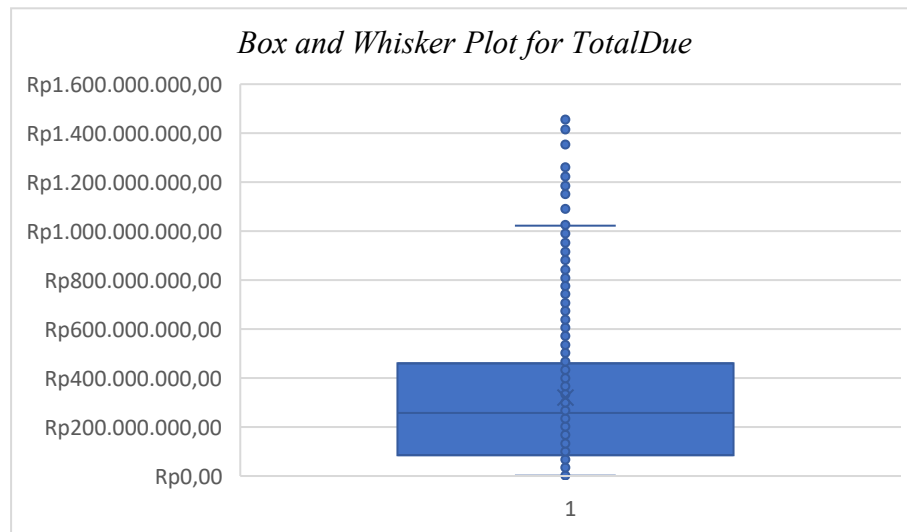


Explanation :

ShippingCost shows a wide spread with no significant outliers, indicating most transactions incur relatively higher shipping costs. The

presence of low shipping costs alongside higher dominant values indicate special cases such as nearby delivery or promo shipping rates.

- Data field TotalDue



Box and Whisker Plot for TotalDue

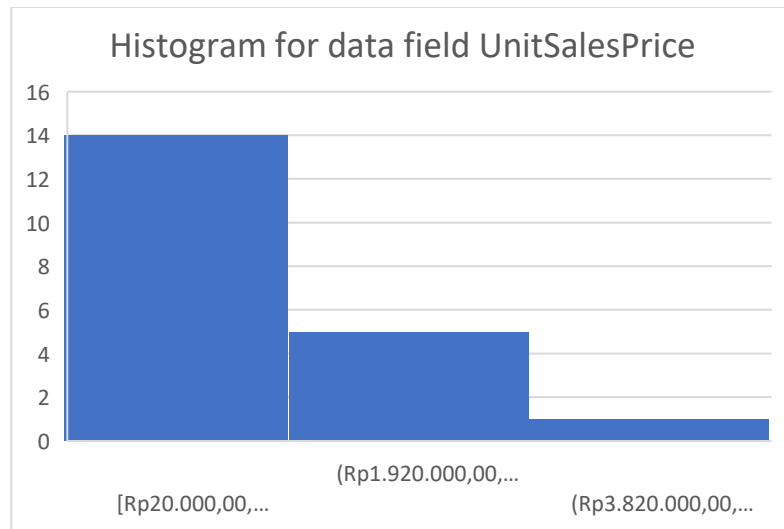
Explanation :

A small number of high-value transactions significantly drive total revenue, suggesting the presence of bulk orders or premium transactions. The concentration of outliers suggests that revenue is not evenly distributed and may depend heavily on a few large transactions.

2. Check skewness and data distribution

2.A Master Data Product

- Data Field UnitSalesPrice

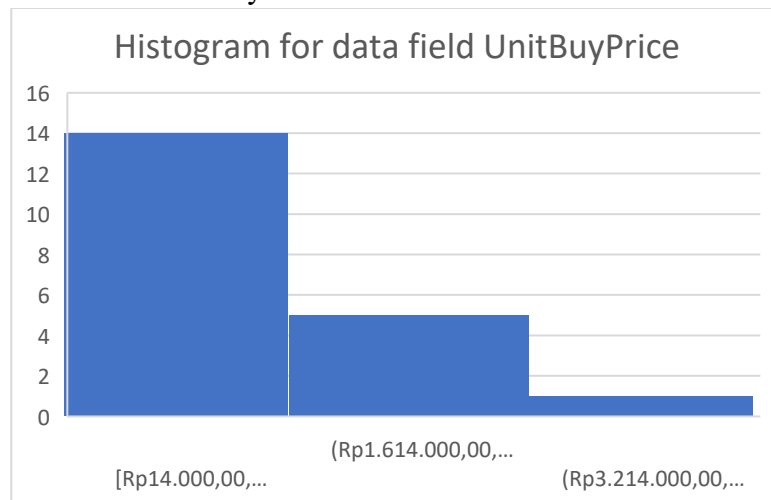


Histogram for data field UnitSalesPrice

Explanation:

The distribution indicates that the product catalog is dominated by lower-priced items, with only a small number of premium products contributing to the higher price range.

- Data Field UnitBuyPrice

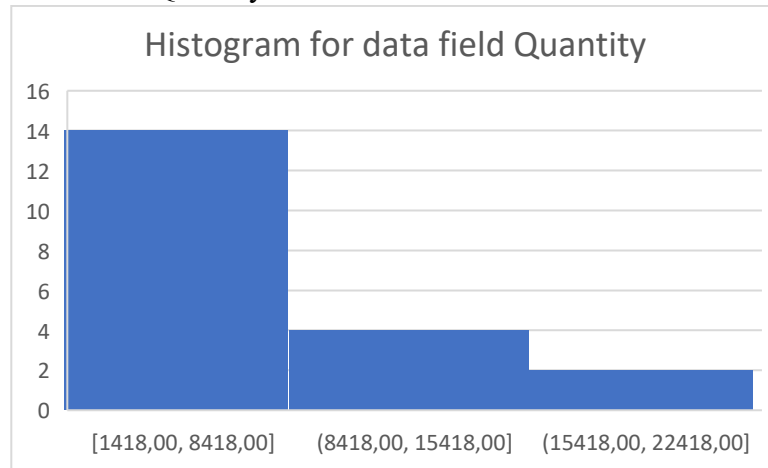


Histogram for data field UnitBuyPrice

Explanation:

The distribution indicates that the product catalog is dominated by lower-priced items, with only a small number of premium products contributing to the higher price range.

- Data Field Quantity

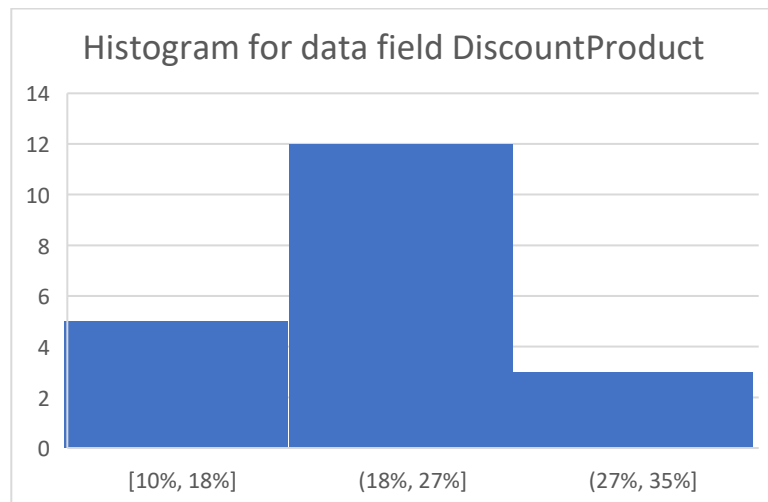


Histogram for data field Quantity

Explanation:

Quantity data is right-skewed, with most products having lower stock levels and a few products showing significantly higher quantities.

- Data Field DiscountProduct



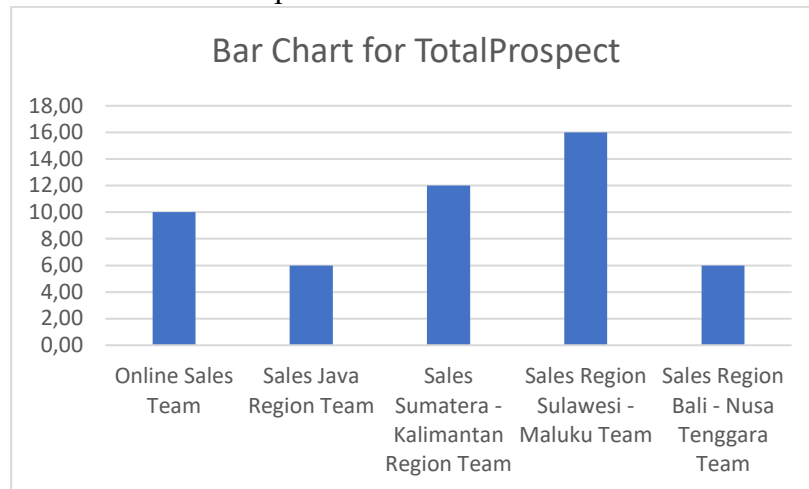
Histogram for data field DiscountProduct

Explanation:

DiscountProduct is consistently distributed within a narrow range, indicating standardized discount application with low variability. The limited variation in discount values may reduce flexibility in pricing strategies but ensures consistency across products.

2.B Master Data Sales Team

- Data Field TotalProspect



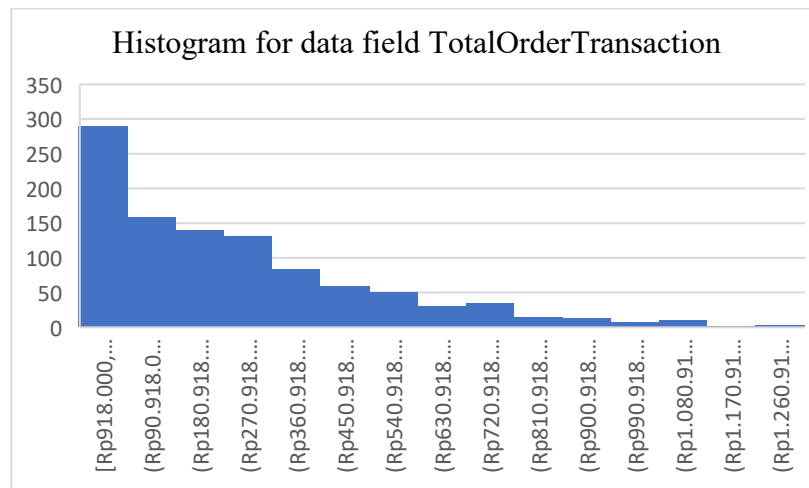
Bar Chart for data field TotalProspect

Explanation:

The higher concentration of prospects in the Sulawesi region may indicate stronger market potential or more aggressive sales activity, while lower values in Java and Bali suggest opportunities for growth or the need for better resource allocation.

2.C Data Sales Order

- Data Field TotalOrderTransaction



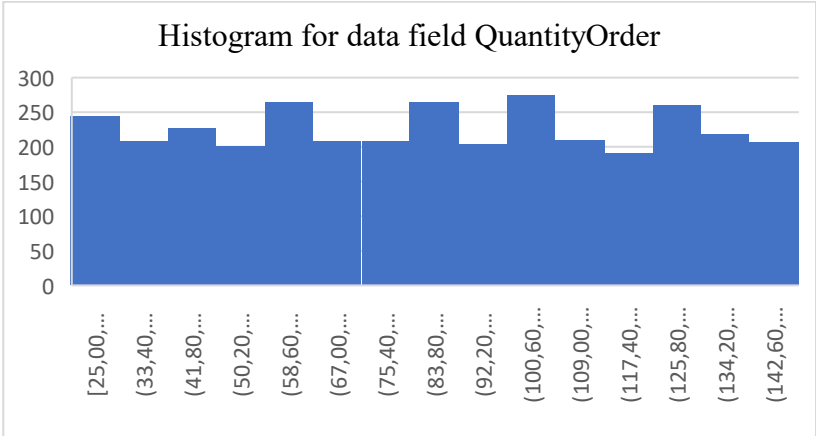
Histogram for data field TotalOrderTransaction

Explanation:

The transaction data follows a Power Law / Long-Tail distribution. Roughly 60–70% of all orders are valued below Rp270M. While the operational focus remains on high-volume processing for smaller orders, the high-value outliers indicate a niche but high-impact segment of premium buyers that requires a different account management strategy.

2.D Data Order Item

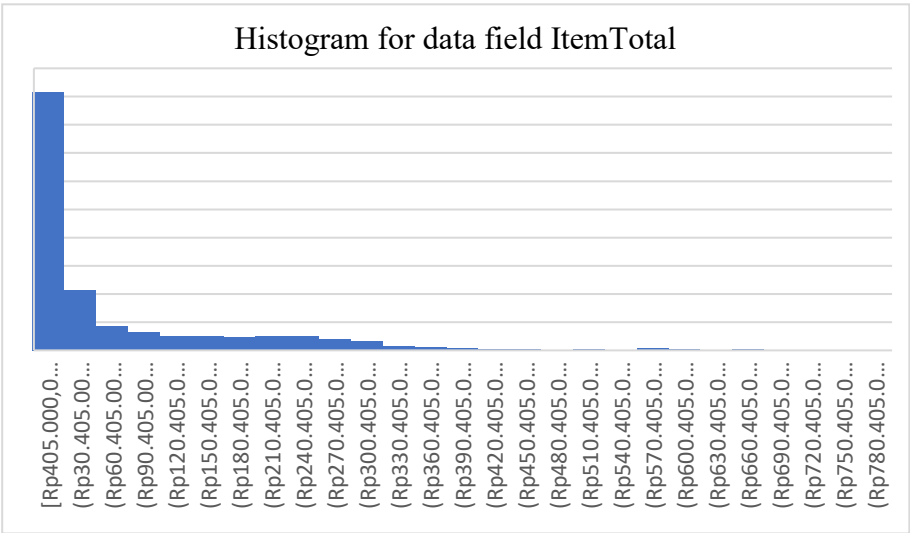
- Data Field QuantityOrder



Histogram for data field QuantityOrder

Explanation:
The QuantityOrder data follows a Uniform Distribution, indicating that order sizes are distributed evenly between 25 and 151 units. This lack of a dominant 'average' order size suggests a highly diverse purchasing behavior. From an operational standpoint, this requires high flexibility in fulfillment and logistics, as there is no predictable 'standard' order volume per transaction.

- Data Field ItemTotal



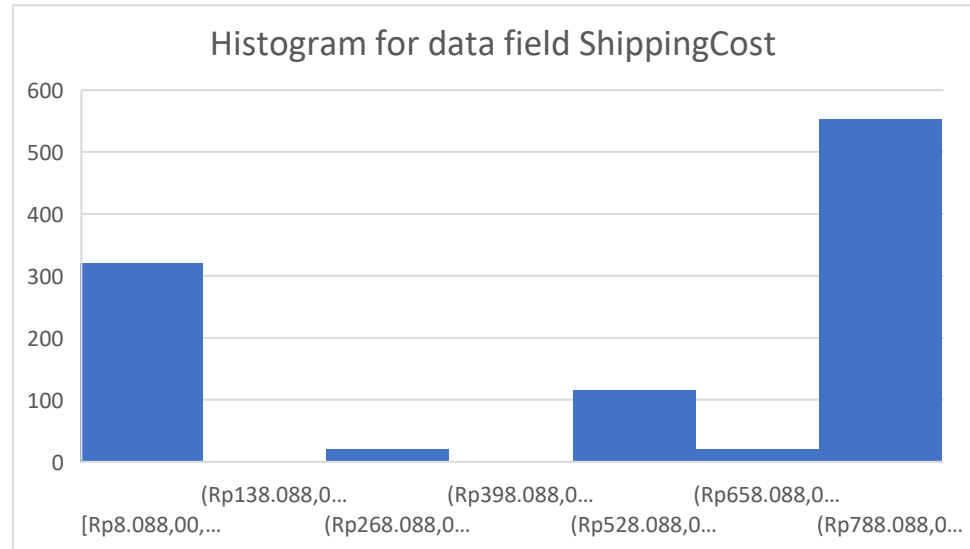
Histogram for data field ItemTotal

Explanation:
The ItemTotal data follows an Extreme Right-Skewed (L-shaped) distribution. Over 75% of all items sold fall within the lowest price bracket. This indicates a high-volume, low-margin operational profile.

While there are extreme outliers in the Rp500M+ range, they represent less than 1% of transaction frequency and should be analyzed separately from the core business activity.

2.E Data Invoice

- Data Field ShippingCost

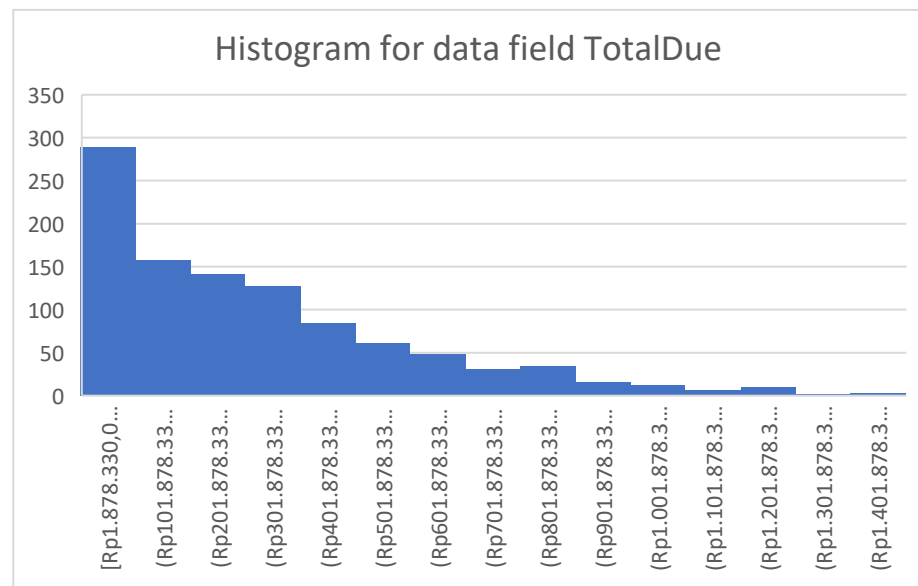


Histogram for data field ShippingCost

Explanation:

ShippingCost shows a clustered distribution pattern, indicating that costs are likely assigned based on predefined tiers rather than continuous calculation.

- Data Field TotalDue



Histogram for data field TotalDue

Explanation:

The histogram shows a strongly right-skewed distribution, where most transactions are concentrated at lower values, while a small number of transactions reach very high amounts. This indicates that a few high-value transactions contribute significantly to overall revenue.

3.5 Data Cleaning file Version Control

Data :

- Raw : Dataset after extracted and merge - First Version.xlsx
 - Original dataset used as the source of truth(no modification)
- Clean : Dataset - Start Data Cleaning.xlsx
 - Processed dataset after data cleaning steps. Each version reflects improvements or corrections
- Test after Clean : Dataset - After Data Cleaning(Testing After Cleaning).xlsx
 - Dataset used for validation and testing after cleaning

Docs :

- Documentation : Data Cleaning Documentation
 - Contain detailed steps, assumptions, rules applied during cleaning process and detail for each data field, data type, and business rules.